

PAVAN DHARMA ADAPA

Clinical Data Coordinator · Clinical Data Analyst

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PROFESSIONAL SUMMARY

M.S. in Computer Science graduate focused on clinical research data. As a graduate research assistant, cleaned and analyzed a 900,000+ record livestock-disease outbreak dataset for Computer Science and Veterinary Medicine faculty. Contributed 36 merged pull requests to a lab's open-source DNA sequencing pipeline. Analyzes public CMS claims filings and CDC survey microdata in SQL and Python.

TECHNICAL SKILLS

Clinical & Health Data: Quality measures (HEDIS, CMS eCQM), value sets, SNOMED CT, LOINC, synthetic EHR data (Synthea), de-identified inpatient encounters (UCI)

Data Management: Data cleaning, quality control and validation, versioned measure specifications, technical documentation and user guides, reproducible pipelines

Analysis & Reporting: SQL, Python (pandas, NumPy, SciPy), R, Excel, report preparation, Git and version control

PROFESSIONAL EXPERIENCE

Data Analytics Intern | *Agricultural Microbiomes Lab, Mississippi State University, Starkville, MS* Jul 2026 – Present

- Contributed **36 merged pull requests** to the lab's open-source DNA sequencing pipeline (maintained by Li Zhang), making it a documented, tested tool that installs in one command and runs on Linux, macOS, WSL2 and Windows.
- Validated pipeline output against an independently built implementation, **agreeing to within 0.01 percentage points** on every species and genus.
- Wrote failure diagnostics that name the real cause instead of a misleading error, and a **1,494-line user guide** that takes a new user from an unconfigured machine through installation and results.

Graduate Teaching Assistant | *Mississippi State University, Starkville, MS* Sep 2025 – May 2026

- Taught data cleaning, transformation and visualization in Python and R to **100+ students across six courses**.

Graduate Research Assistant | *Mississippi State University, Starkville, MS* Aug 2024 – Aug 2025

- Cleaned and analyzed a **900,000+ record** Vesicular Stomatitis Virus (VSV) livestock-outbreak dataset, and presented the findings to Computer Science and Veterinary Medicine faculty.
- Co-author (third of five) on a review manuscript under revision after peer review; contributed the data preparation, statistical modeling and results reporting, and drafted technical sections.
- Built a Python/Selenium scraper that collected genome metadata for a poultry respiratory-disease project, **cutting collection from hours to under 10 minutes per dataset**.

HEALTH DATA PROJECTS

Clinical Quality Measures Engine — *HEDIS / CMS eCQM · synthetic non-PHI EHR (Synthea)*

- Computed HEDIS and CMS eCQM measures over a **13,810-patient EHR** (820,993 encounters, 10.8M observations), holding each measure as a versioned specification with its SNOMED CT and LOINC value sets separate from the engine — so an annual steward update is a reviewable diff a clinical analyst signs off, not a code change.
- Traced **203 of 204** diabetes-measure failures to missing HbA1c tests, an artifact of how the Synthea generator schedules labs, and reported it as an engine check rather than a finding about care.

30-Day Readmission Risk — *UCI Diabetes 130-US Hospitals, de-identified*

- Modeled **99,343 real inpatient encounters**: working the top 20% of the risk list captures **37.7% of all readmissions** at 1.89× enrichment, number-needed-to-screen 4.8.
- Reported AUC 0.664 and argued the ranking — not the AUC — is what a capacity-constrained care-management team consumes.

EDUCATION

M.S., Computer Science — Mississippi State University, Mississippi State, MS May 2026

B.Tech, Computer Science — Malla Reddy University, Hyderabad, India May 2024

Publication (under revision): Li Zhang, Saikanth Ratnavale, Pavan Dharma Adapa, Sai Harika Gade, Michael E. Navicky — Computational Tools for Modeling and Predicting Respiratory Disease Spread in Commercial Poultry Production.